Classification Accuracy

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~~1. Introduction~~

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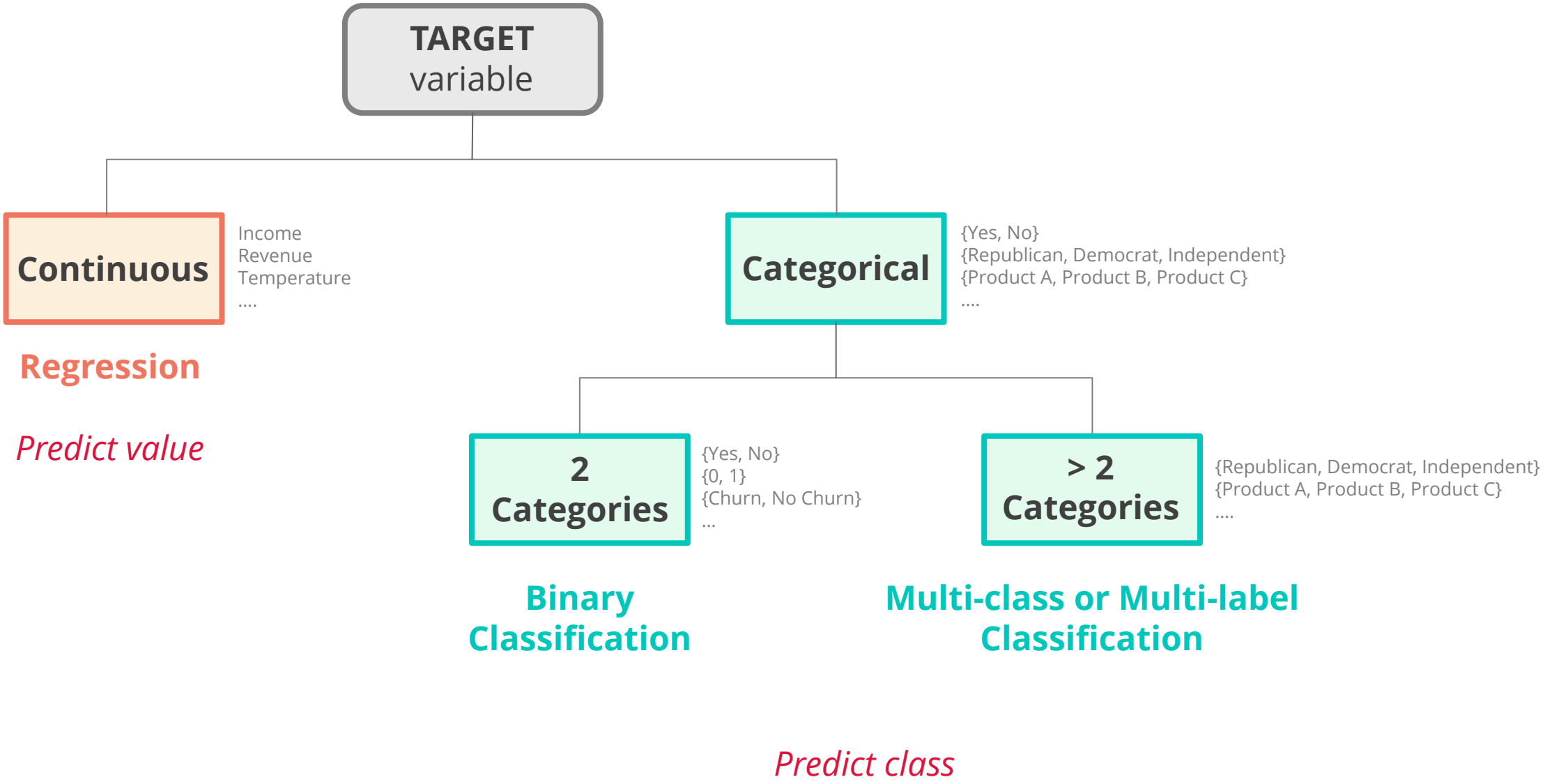

TARGET variable
Continuous
Income
Revenue
Temperature
....
Regression
Predict value
Categorical
{Yes, No}
{Republican, Democrat, Independent}
{Product A, Product B, Product C}
....
2 Categories
{Yes, No}
{0, 1}
{Churn, No Churn}
...
Binary Classification
> 2 Categories
{Republican, Democrat, Independent}
{Product A, Product B, Product C}
....
Multi-class or Multi-label Classification
Predict class

Regression Models

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \dots & x_{1j} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2j} \\ x_{31} & x_{32} & x_{33} & \dots & x_{3j} \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ x_{n1} & x_{n2} & x_{n3} & \dots & x_{nj} \end{pmatrix} \quad y = \begin{pmatrix} 40 \\ 10 \\ 0 \\ . \\ . \\ . \\ 50 \end{pmatrix} \quad \hat{y} = \begin{pmatrix} 50 \\ 10 \\ 10 \\ . \\ . \\ . \\ 70 \end{pmatrix} \quad \varepsilon = \begin{pmatrix} 10 \\ 0 \\ -10 \\ . \\ . \\ . \\ 20 \end{pmatrix}$$

Classification Models

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \dots & x_{1j} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2j} \\ x_{31} & x_{32} & x_{33} & \dots & x_{3j} \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ x_{n1} & x_{n2} & x_{n3} & \dots & x_{nj} \end{pmatrix} \quad y = \begin{pmatrix} 1 \\ 1 \\ 0 \\ . \\ . \\ . \\ 0 \end{pmatrix} \quad \hat{y} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$$

?

Classification Models

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \dots & x_{1j} \\ x_{21} & x_{22} & x_{23} & \dots & x_{2j} \\ x_{31} & x_{32} & x_{33} & \dots & x_{3j} \\ . & . & . & & . \\ . & . & . & & . \\ . & . & . & & . \\ x_{n1} & x_{n2} & x_{n3} & \dots & x_{nj} \end{pmatrix} \quad y = \begin{pmatrix} 1 \\ 1 \\ 0 \\ . \\ . \\ . \\ 0 \end{pmatrix} \quad \hat{p} = \begin{pmatrix} 0.29 \\ 0.90 \\ 0.31 \\ . \\ . \\ . \\ 0.47 \end{pmatrix} \quad \hat{y} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ . \\ . \\ . \\ 0 \end{pmatrix}$$

`predict_proba()` `predict()`

True Class (y)	Predicted Probability ($\hat{p}$)
+	0.29
+	0.90
−	0.31
−	0.89
+	0.95
+	0.71
−	0.24
+	0.12
−	0.59
−	0.47

True Class (y)	Predicted Probability ($\hat{p}$)
+	0.95
+	0.90
−	0.89
+	0.71
−	0.59
−	0.47
−	0.31
+	0.29
−	0.24
+	0.12

True Class (y)	Predicted Probability ($\hat{p}$) ↓	Predicted Class	True / False?	True Positive	False Positive	True Negative	False Negative
+	0.95	+	True Positive	1	0	0	0
+	0.90	+	True Positive	2	0	0	0
−	0.89	+	False Positive	2	1	0	0
+	0.71	+	True Positive	3	1	0	0
−	0.59	+	False Positive	3	2	0	0
−	0.47	−	True Negative	3	2	1	0
−	0.31	−	True Negative	3	2	2	0
+	0.29	−	False Negative	3	2	2	1
−	0.24	−	True Negative	3	2	3	0
+	0.12	−	False Negative	**3**	**2**	**3**	**2**

True Class (y)	Predicted Probability ($\hat{p}$) ↓	Predicted Class	True / False?
+	0.95	+	True Positive
+	0.90	+	True Positive
–	0.89	+	False Positive
+	0.71	+	True Positive
–	0.59	+	False Positive
–	0.47	–	True Negative
–	0.31	–	True Negative
+	0.29	–	False Negative
–	0.24	–	True Negative
+	0.12	–	False Negative

True Positive	False Positive	True Negative	False Negative
3	2	3	2

		Predicted Class	
		+	–
True Class	+	True Positive 3	False Negative 2
	–	False Positive 2	True Negative 3

Confusion matrix

True Class (y)	Predicted Probability ($\hat{p}$) ↓
+	0.95
+	0.90
−	0.89
+	0.71
−	0.59
−	0.47
−	0.31
+	0.29
−	0.24
+	0.12

True Positive	False Positive	True Negative	False Negative
3	2	3	2

ACCURACY $= \dfrac{\text{TP} + \text{TN}}{\text{Total Population}} = \dfrac{3+3}{10} = 60.0\%$

SENSITIVITY / **RECALL** $= \dfrac{\text{TP}}{\text{Total Positives}} = \dfrac{3}{3+2} = 60.0\%$

True Positive Rate

SPECIFICITY $= \dfrac{\text{TN}}{\text{Total Negatives}} = \dfrac{3}{3+2} = 60.0\%$

True Negative Rate

$$\text{ACCURACY} = \frac{TP + TN}{\text{Total Population}}$$

- Imagine disease prevalence: 1 in 100.
- A model predicts ‘healthy’ (negative) for everyone.
- What’s the **accuracy** of this model?
- 99%

Accuracy is often the most misleading metric in classification!

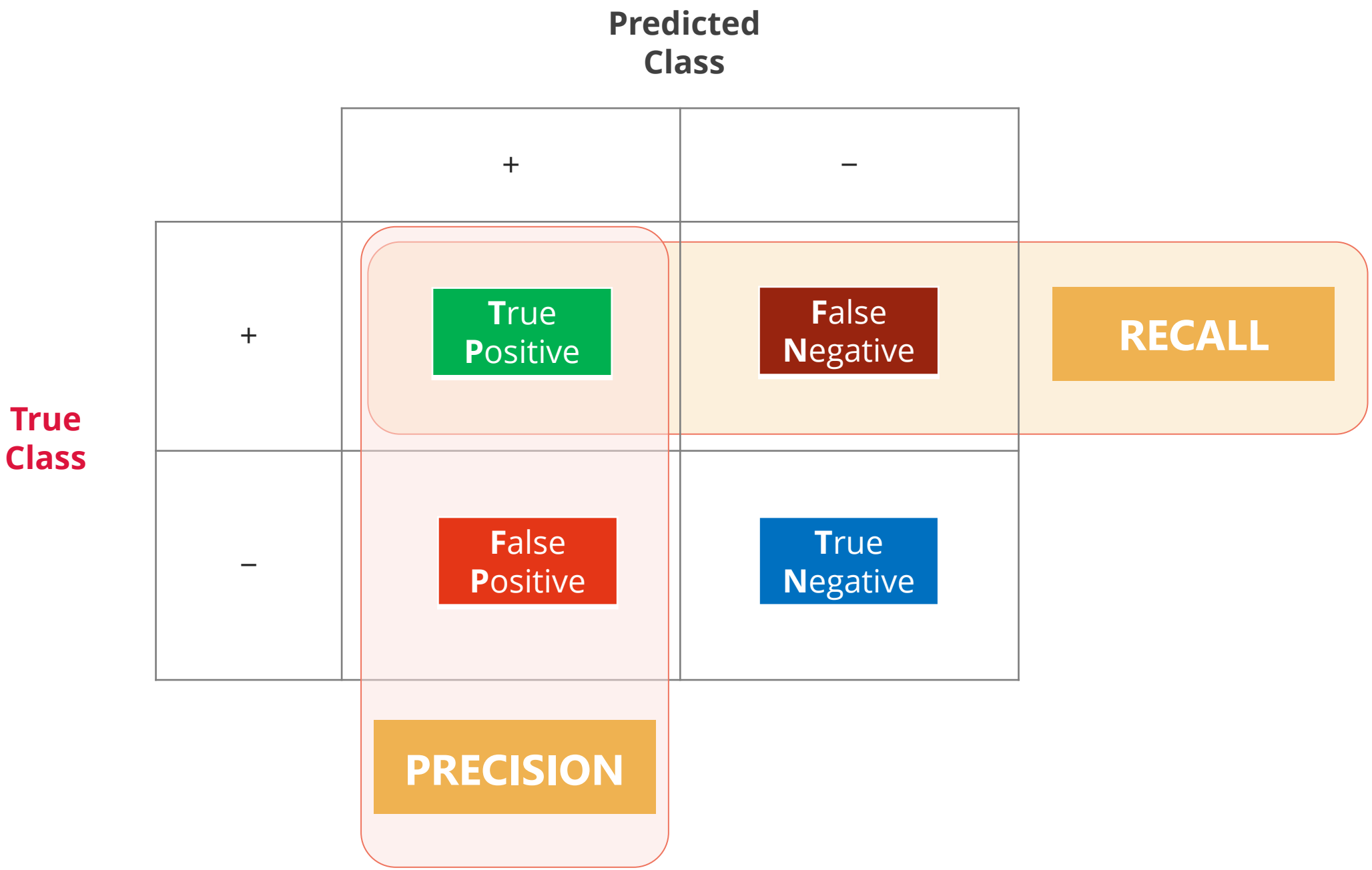
True Class (y)	Predicted Probability ($\hat{p}$) ↓
+	0.95
+	0.90
−	0.89
+	0.71
−	0.59
−	0.47
−	0.31
+	0.29
−	0.24
+	0.12

True Positive	False Positive	True Negative	False Negative
3	2	3	2

RECALL $= \dfrac{\text{TP}}{\text{Total Positives}} = 60.0\%$

PRECISION $= \dfrac{\text{TP}}{\text{Predicted Positives}} = \dfrac{3}{3+2} = 60.0\%$

F1 SCORE $= \dfrac{1}{\dfrac{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}{2}} = \dfrac{1}{\dfrac{\frac{1}{0.6} + \frac{1}{0.6}}{2}} = 0.60$


Predicted Class
+
−
True Class
+
−
True Positive
False Negative
RECALL
False Positive
True Negative
PRECISION

True Class (y)	Predicted Probability ($\hat{p}$) ↓	Predicted Class
+	0.95	+
+	0.90	+
−	0.89	+
+	0.71	+
−	0.59	+
−	0.47	−
−	0.31	−
+	0.29	−
−	0.24	−
+	0.12	−

THRESHOLD = **0.5**

True Positive	False Positive	True Negative	False Negative
3	2	3	2

Measure	Value
Accuracy	0.60
Sensitivity (Recall)	0.60
Specificity	0.60
Precision	0.60
F1 Score	0.60

True Class (y)	Predicted Probability ($\hat{p}$) ↓	Predicted Class
+	0.95	+
+	0.90	+
−	0.89	+
+	0.71	+
−	0.59	−
−	0.47	−
−	0.31	−
+	0.29	−
−	0.24	−
+	0.12	−

THRESHOLD = **0.6**

True Positive	False Positive	True Negative	False Negative
3	1	4	2

Measure	Value
Accuracy	0.70
Sensitivity (Recall)	0.60
Specificity	0.80
Precision	0.75
F1 Score	0.67

PROBABILITY THRESHOLD VALUES

Measure	0.1	0.2	0.3	0.4	**0.5**	0.6	0.7	0.8	0.9
Accuracy	0.50	0.40	0.40	0.50	0.60	**0.70**	**0.70**	0.60	**0.70**
Sensitivity (Recall)	**1.00**	0.80	0.60	0.60	0.60	0.60	0.60	0.40	0.40
Specificity	--	--	0.20	0.40	0.60	0.80	0.80	0.80	**1.00**
Precision	0.50	0.44	0.43	0.50	0.60	0.75	0.75	0.67	**1.00**
F1 Score	**0.67**	0.57	0.50	0.55	0.60	**0.67**	**0.67**	0.50	0.57

1. The values depend on the decision boundary.

Breathalyzers

ASSUMPTIONS:

1. **One in a thousand** drivers is driving drunk.
2. A breathalyzer **never** fails to detect a truly drunk person.
3. However, it displays false drunkenness in **5%** of the cases when the driver is actually sober.
4. A police officer stops a **random** driver and asks her to take the breathalyzer test.
5. The test indicates that the driver **is** drunk.

What is the probability that the driver is DUI?

(a) 95% **(b)** 90% **(c)** 5% **(d)** 2%

1,000 Tests

One Actual Drunk Driver

The test never fails to detect a truly drunk person.
Hence, the test result for this person would display drunkenness (aka "positive").

True Positive	False Positive
1	0

999 Responsible Drivers

The tests would show false drunkenness in 5% of these cases.
Hence, ~50 of these tests would display drunkenness (aka "positives").

True Positive	False Positive
0	50

PRECISION $= \dfrac{\text{TP}}{\text{Predicted Positives}} = \dfrac{1+0}{1+0+50+0} \approx 2.0\%$

This is the probability that *one* of the drivers among all drivers who tested positive is actually drunk.

ALTERNATIVE SCENARIO:

1,000 Tests

500
Drunk Drivers

The test never fails to detect a truly drunk person.
Hence, the test result for these drivers would display drunkenness (aka "positives").

True Positive	False Positive
500	0

500
Responsible Drivers

The tests would show false drunkenness in 5% of these cases.
Hence, 25 test results would display drunkenness (aka "positives").

True Positive	False Positive
0	25

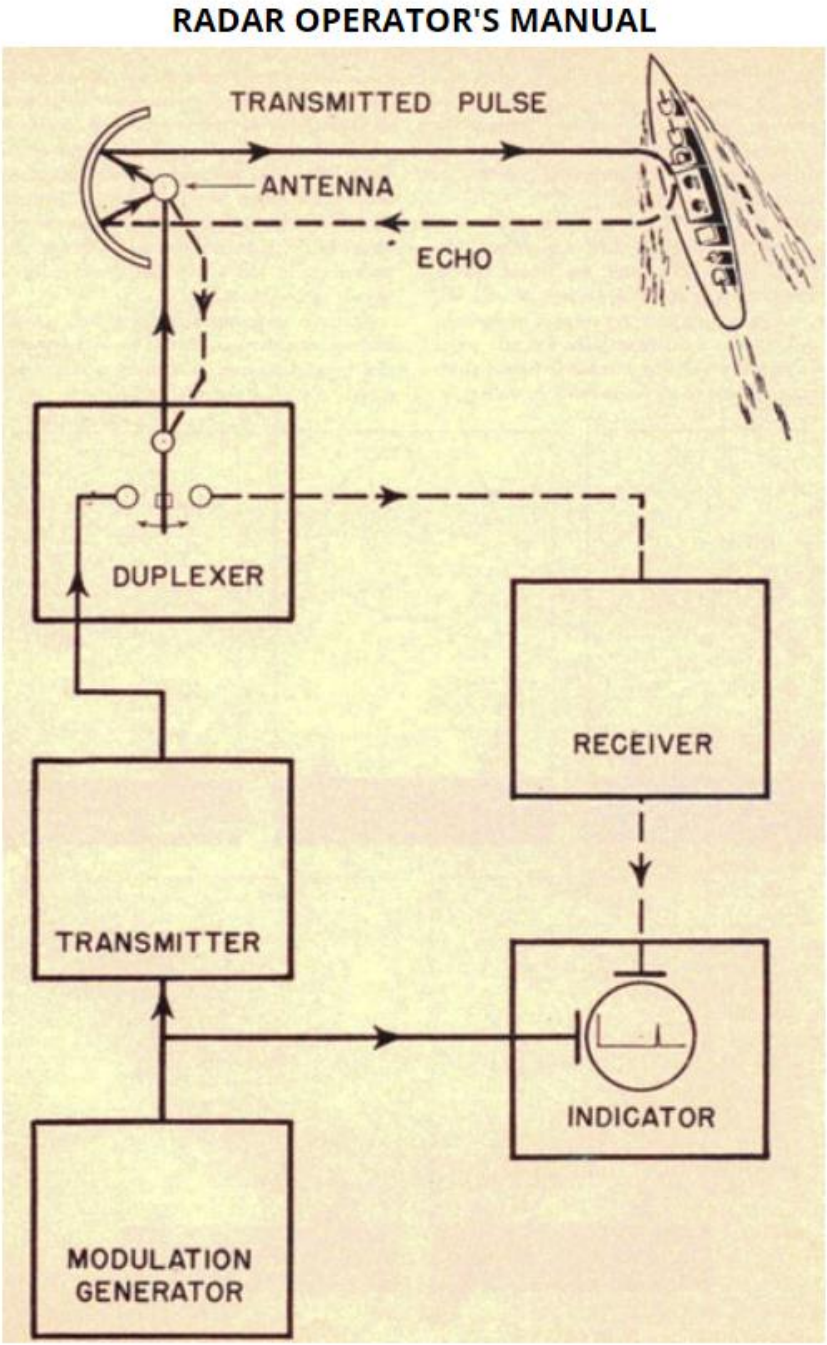
PRECISION $= \dfrac{\text{TP}}{\text{Predicted Positives}} = \dfrac{500}{500 + 25} \approx 95.0\%$

This is the probability that *one* of the drivers among all drivers who tested positive is actually drunk.

1. The values depend on the **decision boundary**.

2. The values depend on the **class probabilities**.

We need a measure that is independent of those two factors.



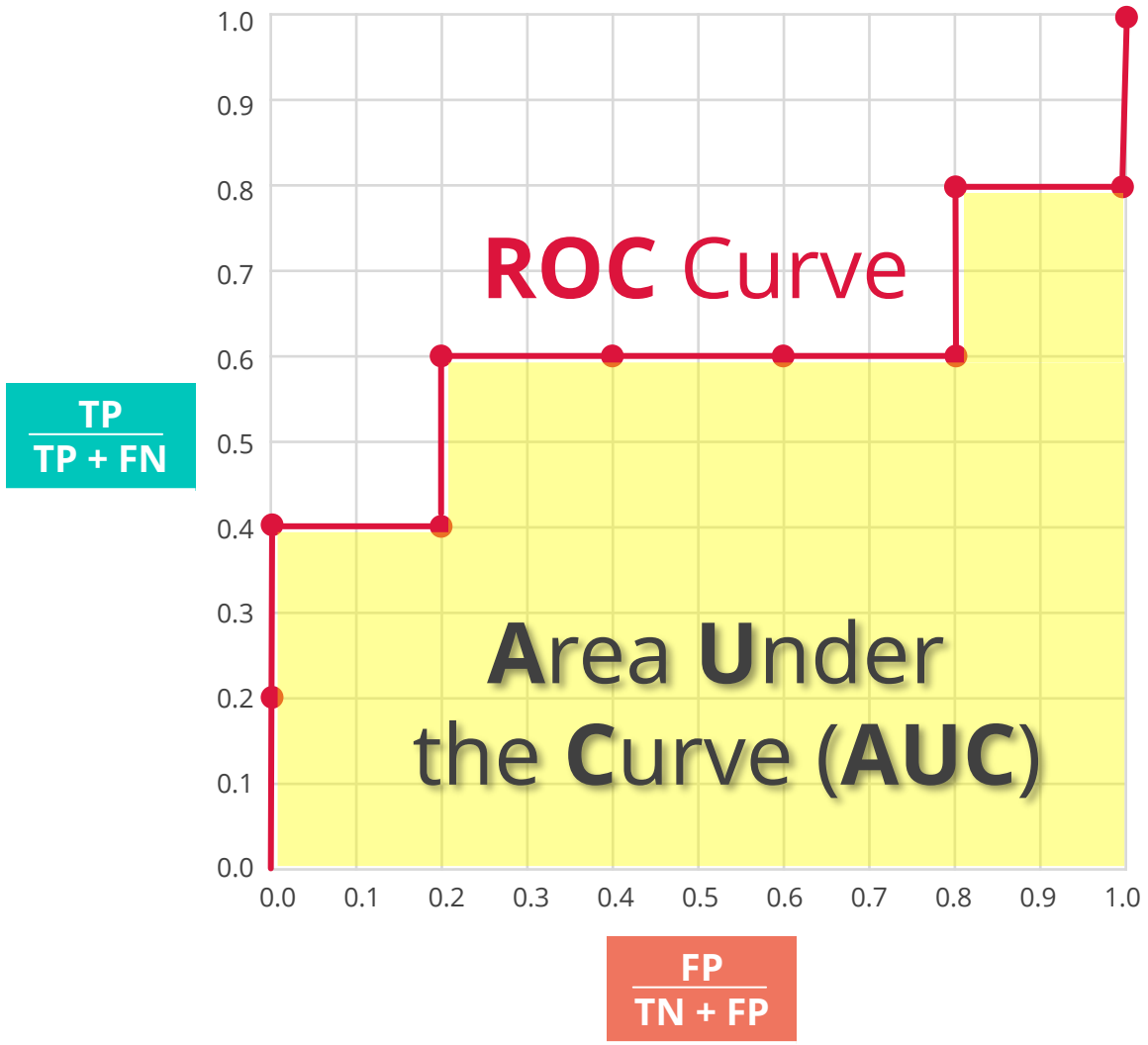
- Following the attack on Pearl Harbor in 1941, the United States army began new research to improve the **predictive accuracy** to correctly detect an enemy aircraft from their radar signals.
- They needed **a new measure** to assess various **radar receiver operators** on their ability to discriminate signal (e.g., enemy aircraft) from noise (e.g., flocks of birds).
- This is a **binary classification** problem!

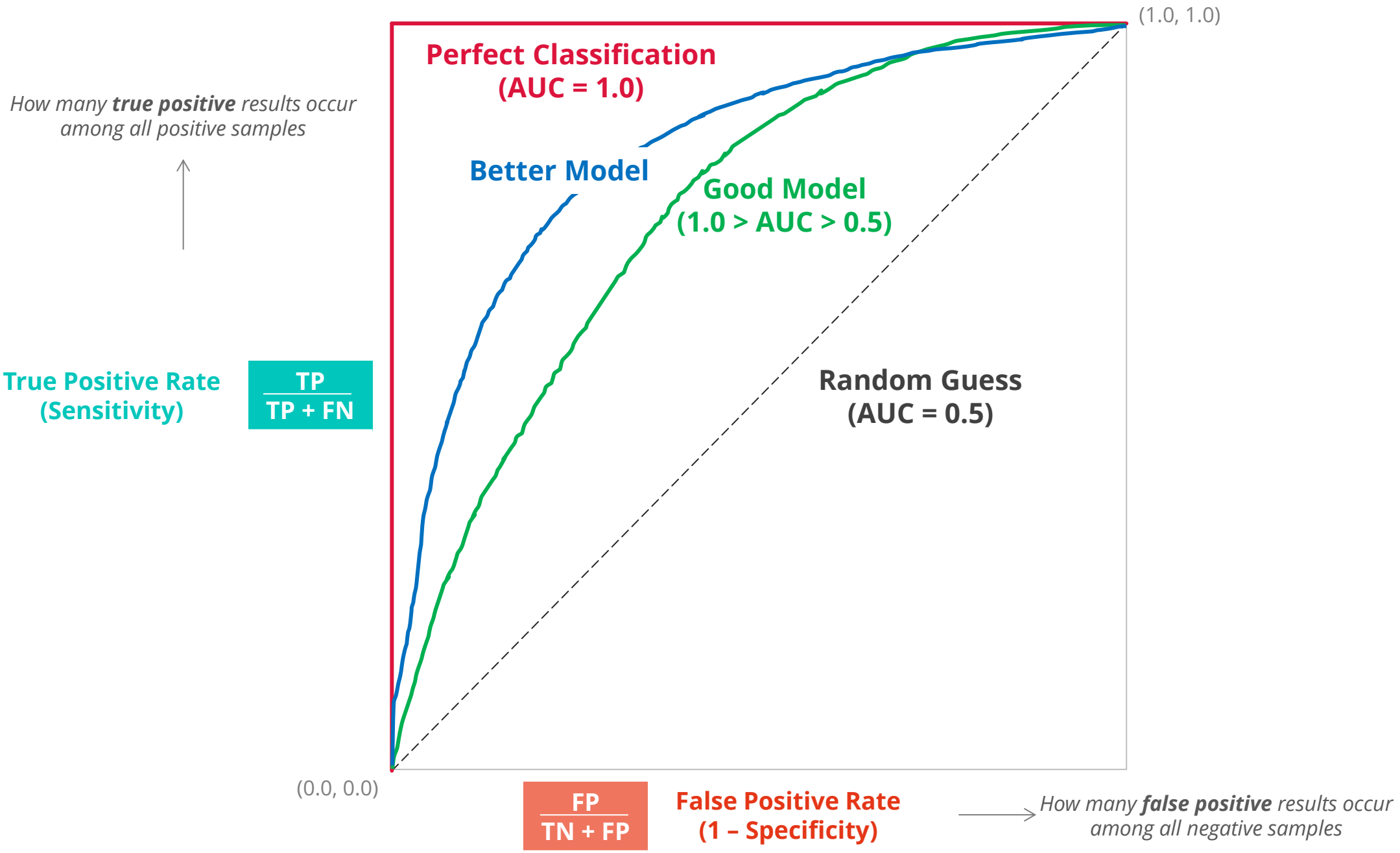
Receiver **O**perating **C**haracteristic

ROC Curve

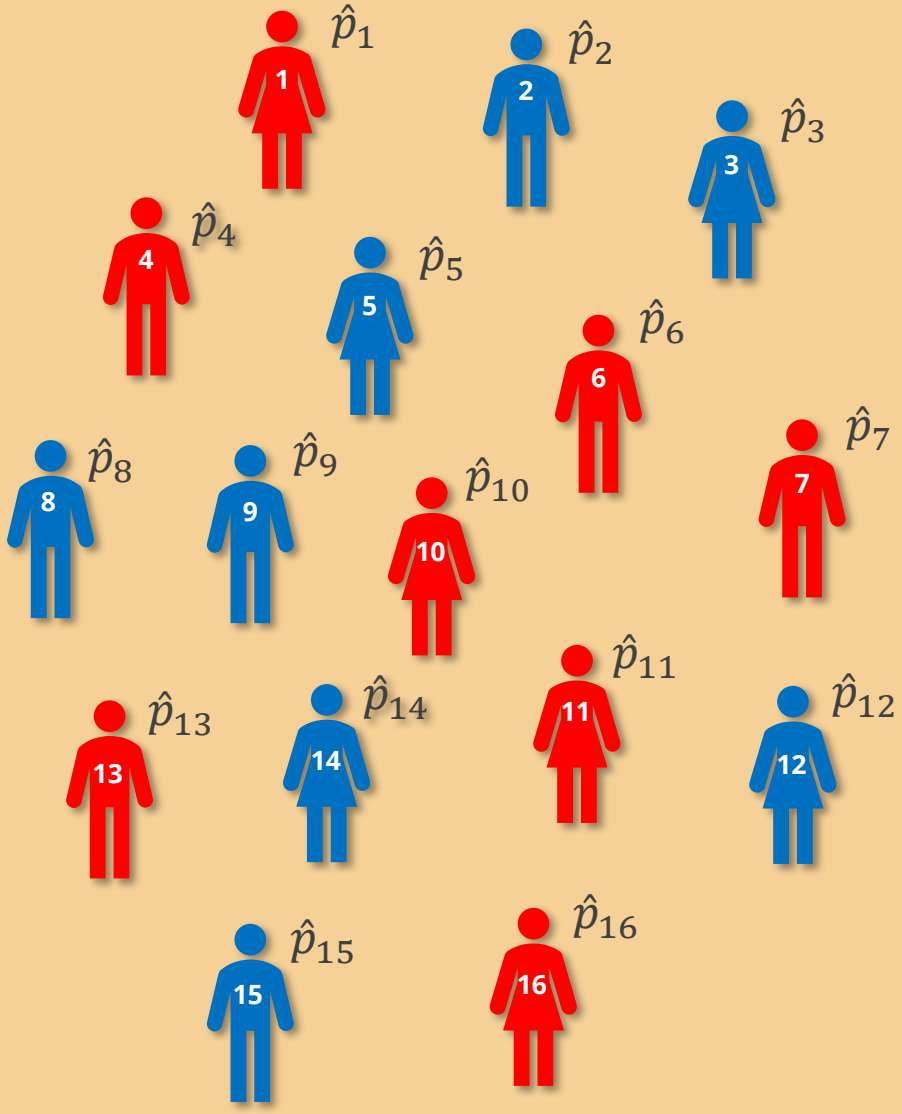
ROC Curve

True Class (y)	Predicted Probability ($\hat{p}$) ↓	$\frac{TP}{TP + FN}$	$\frac{FP}{TN + FP}$
+	0.95	0.2	0.0
+	0.90	0.4	0.0
–	0.89	0.4	0.2
+	0.71	0.6	0.2
–	0.59	0.6	0.4
–	0.47	0.6	0.6
–	0.31	0.6	0.8
+	0.29	0.8	0.8
–	0.24	0.8	1.0
+	0.12	1.0	1.0




(1.0, 1.0)
Perfect Classification
(AUC = 1.0)
How many true positive results occur among all positive samples
Better Model
Good Model
(1.0 > AUC > 0.5)
True Positive Rate
(Sensitivity)
TP / TP + FN
Random Guess
(AUC = 0.5)
(0.0, 0.0)
FP / TN + FP
False Positive Rate
(1 – Specificity)
How many false positive results occur among all negative samples

$\hat{p}$ = **Probability of being a Republican**



Actual Republican **Actual Democrat**

(Both randomly selected)

What is the **probability** that $\hat{p}_4$ is greater than $\hat{p}_{14}$?

↓

AUC

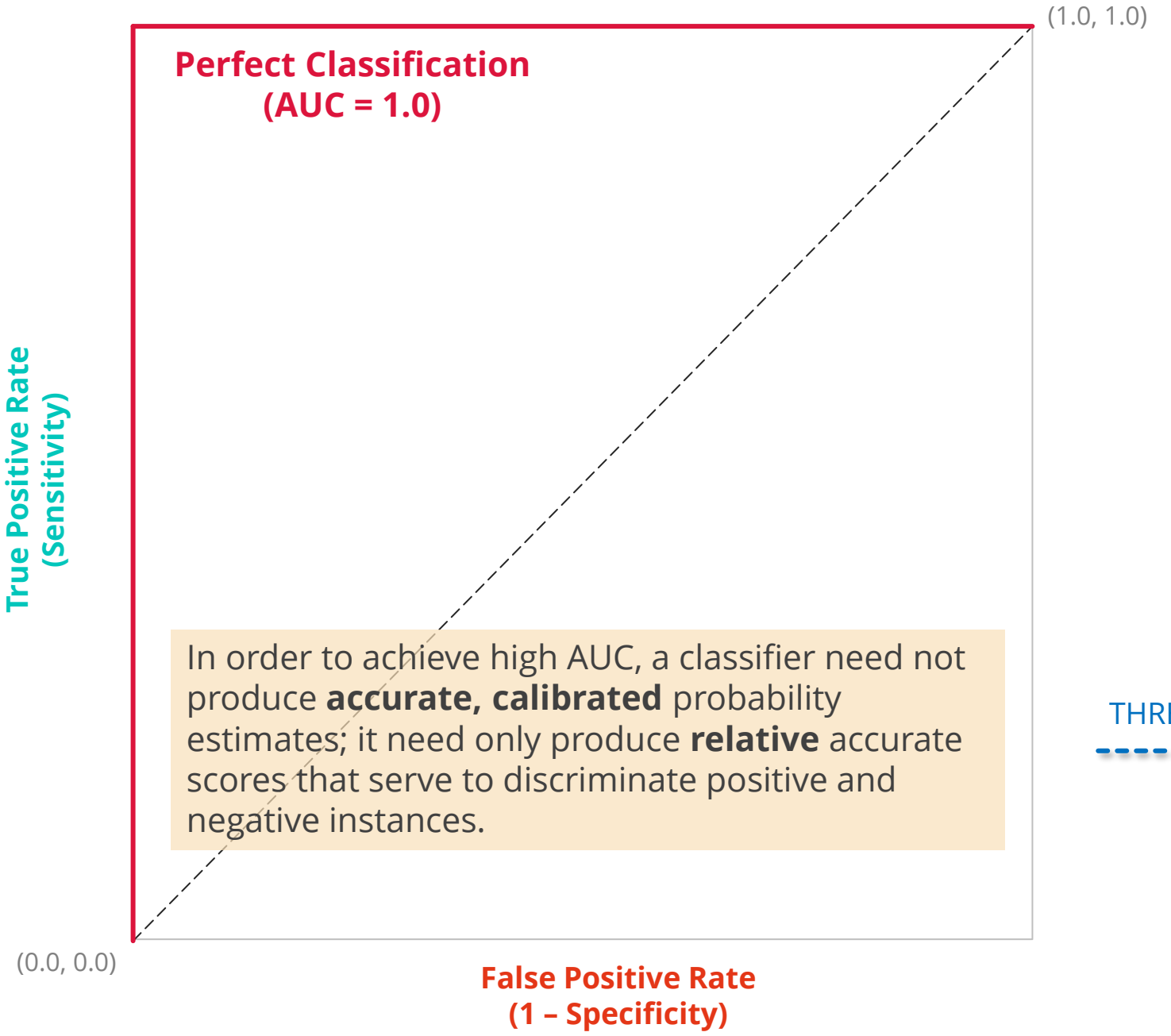
AUC: Probabilistic Interpretation

The area under the curve (AUC) is equal to the probability

that a classifier will rank

a randomly chosen positive instance

higher than a randomly chosen negative one



In order to achieve high AUC, a classifier need not produce **accurate, calibrated** probability estimates; it need only produce **relative** accurate scores that serve to discriminate positive and negative instances.

ACCURACY = 80%!

y	$\hat{p}$	$\hat{y}$
1	0.99999	1
1	0.99999	1
1	0.99993	1
1	0.99986	1
1	0.99964	1
1	0.99955	1
0	0.68139	1
0	0.50961	1
0	0.48880	0
0	0.44951	0

THRESHOLD = **0.5**

http://binf.gmu.edu/mmasso/ROC101.pdf

	MODEL **A**	MODEL **B**
True Class (y)	**Predicted Probability ($\hat{p}$)**	**Predicted Probability ($\hat{p}$)**
+	0.29 (−) ✗	0.11 (−) ✗
+	0.90 (+)	0.90 (+)
−	0.31 (−)	0.31 (−)
−	0.89 (+) ✗	0.98 (+) ✗
+	0.95 (+)	0.95 (+)
+	0.71 (+)	0.71 (+)
−	0.24 (−)	0.24 (−)
+	0.12 (−) ✗	0.02 (−) ✗
−	0.59 (+) ✗	0.90 (+) ✗
−	0.47 (−)	0.47 (−)

Which model is better?

	MODEL **A**	MODEL **B**
True Class (y)	**Predicted Probability ($\hat{p}$)**	**Predicted Probability ($\hat{p}$)**
+	0.29 (−)	0.11 (−)
−	0.89 (+)	0.98 (+)
+	0.12 (−)	0.02 (−)
−	0.59 (+)	0.91 (+)

Equivocally wrong

Emphatically wrong

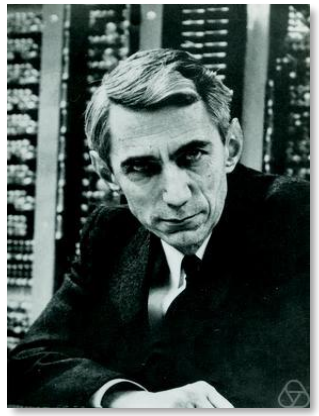
Log Loss / Cross-entropy Loss

$$logLoss = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i(\log(\hat{p}_i)) + (1-y_i)\log(1-\hat{p}_i)\right)$$

If $y_i = 1$ and $\hat{p}_i$ is high → Good!

If $y_i = 0$ and $\hat{p}_i$ is low → Good!

$$H(p,q) = -\sum_i p_i \log q_i$$



If a model were a detective, which metric would best describe its ability to **never miss a clue**, no matter how small?

A. Accuracy

B. Precision

C. Recall

D. Specificity

$$\text{ACCURACY} = \frac{TP + TN}{\text{Total Population}}$$

$$\text{PRECISION} = \frac{TP}{\text{Predicted Positives}}$$

$$\text{RECALL} = \frac{TP}{\text{Total Positives}}$$

$$\text{SPECIFICITY} = \frac{TN}{\text{Total Negatives}}$$

Just like a detective who can't afford to overlook any leads,
recall measures a model's ability to identify **all actual positives**.

Which metric acts like a weather forecast's ability to correctly predict whether it will rain or not, without falsely alarming sunny days?

A. Accuracy

B. Precision

C. Recall

D. Specificity

$$\text{ACCURACY} = \frac{TP + TN}{\text{Total Population}}$$

$$\text{PRECISION} = \frac{TP}{\text{Predicted Positives}}$$

$$\text{RECALL} = \frac{TP}{\text{Total Positives}}$$

$$\text{SPECIFICITY} = \frac{TN}{\text{Total Negatives}}$$

Precision is like predicting rain accurately when it actually rains, avoiding false alarms on sunny days.

If you had to choose a metric to be the judge of a dance competition, ensuring that all good dancers move to the finals without missing any, which one would you choose?

A. Accuracy

B. Precision

C. Recall

D. Specificity

$$\text{ACCURACY} = \frac{TP + TN}{\text{Total Population}}$$

$$\text{PRECISION} = \frac{TP}{\text{Predicted Positives}}$$

$$\text{RECALL} = \frac{TP}{\text{Total Positives}}$$

$$\text{SPECIFICITY} = \frac{TN}{\text{Total Negatives}}$$

Recall ensures **all true positives (talents) are identified** and given their due recognition, just like ensuring all good dancers don't miss out on the finals.

Imagine a model is a goalkeeper. Which metric would best represent its ability to correctly ignore balls that are going out of bounds?

A. Accuracy

B. Precision

C. Recall

D. Specificity

$$\text{ACCURACY} = \frac{TP + TN}{\text{Total Population}}$$

$$\text{PRECISION} = \frac{TP}{\text{Predicted Positives}}$$

$$\text{RECALL} = \frac{TP}{\text{Total Positives}}$$

$$\text{SPECIFICITY} = \frac{TN}{\text{Total Negatives}}$$

Specificity in this context means the goalkeeper is excellent at identifying balls that are not going to enter the goal – aka the **true negatives** – such as shots going wide or high of the goal, thus not mistakenly reacting to non-threats.

If a classification model were a student, which metric would represent its GPA, combining both its precision and recall?

A. Accuracy

B. ROC Curve

C. F1 Score

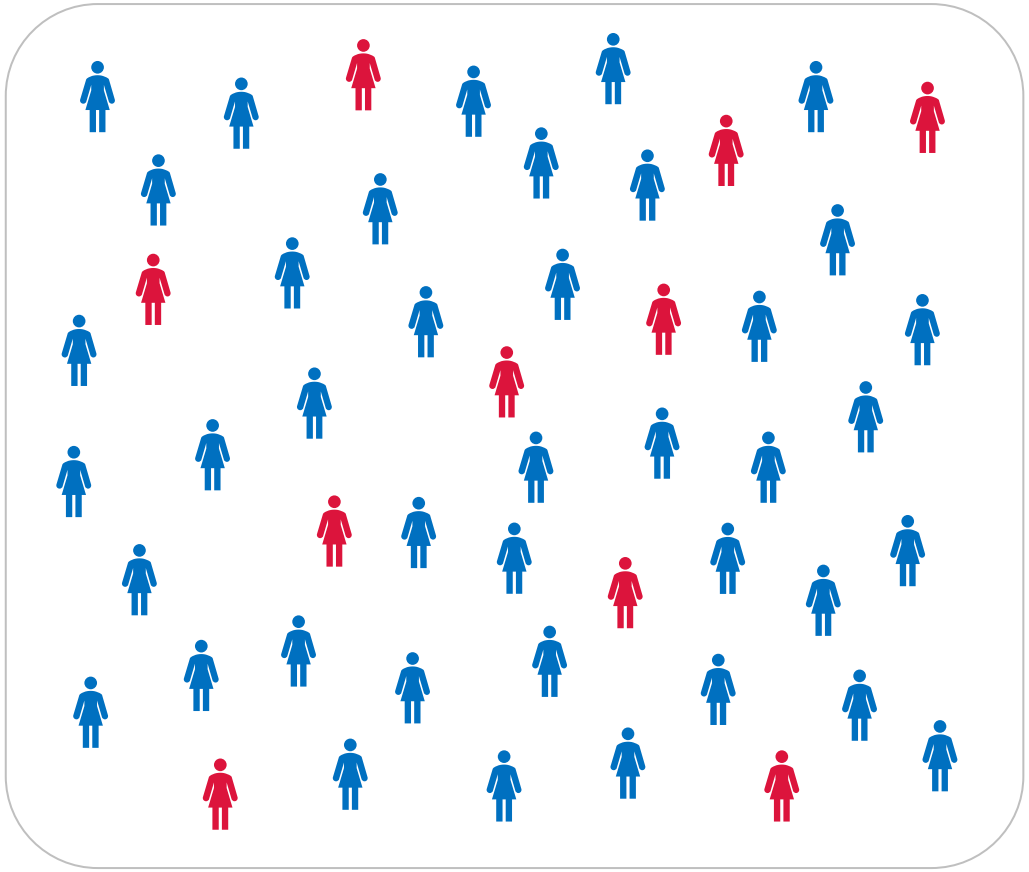
D. AUC

Just like a GPA that combines various aspects of a student's performance, the **F1 Score combines precision and recall** into a single metric.

The boy who cried wolf repeatedly when no wolf exists had _____ **recall** and _____ **precision**.

(he detects every wolf) but terrible precision (lots of false alarms)

Model Lift and Gain



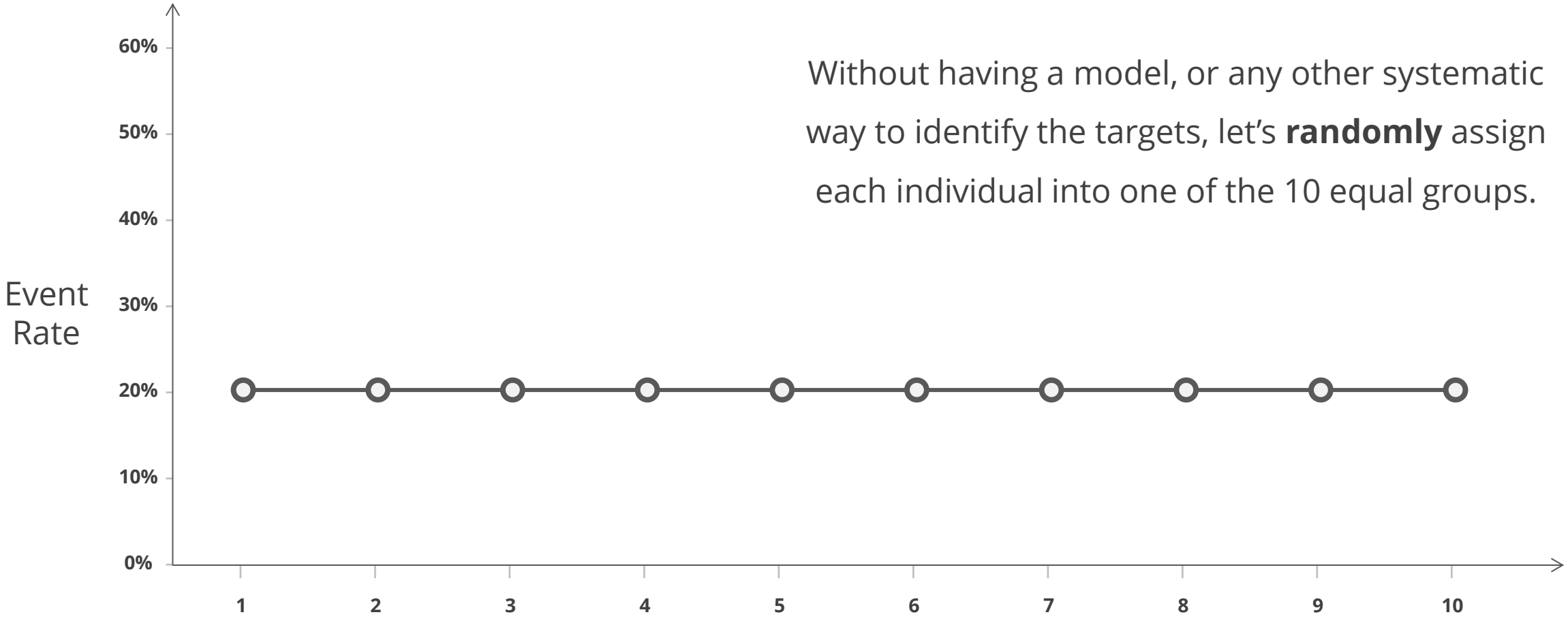
N = 50

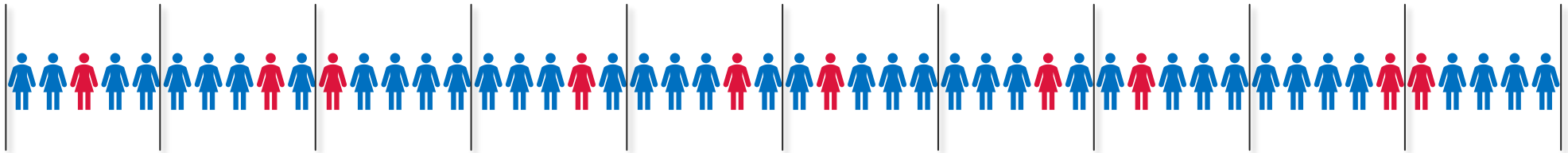
Targets / Events = 10

Non-events = 40

Event Rate = 20%

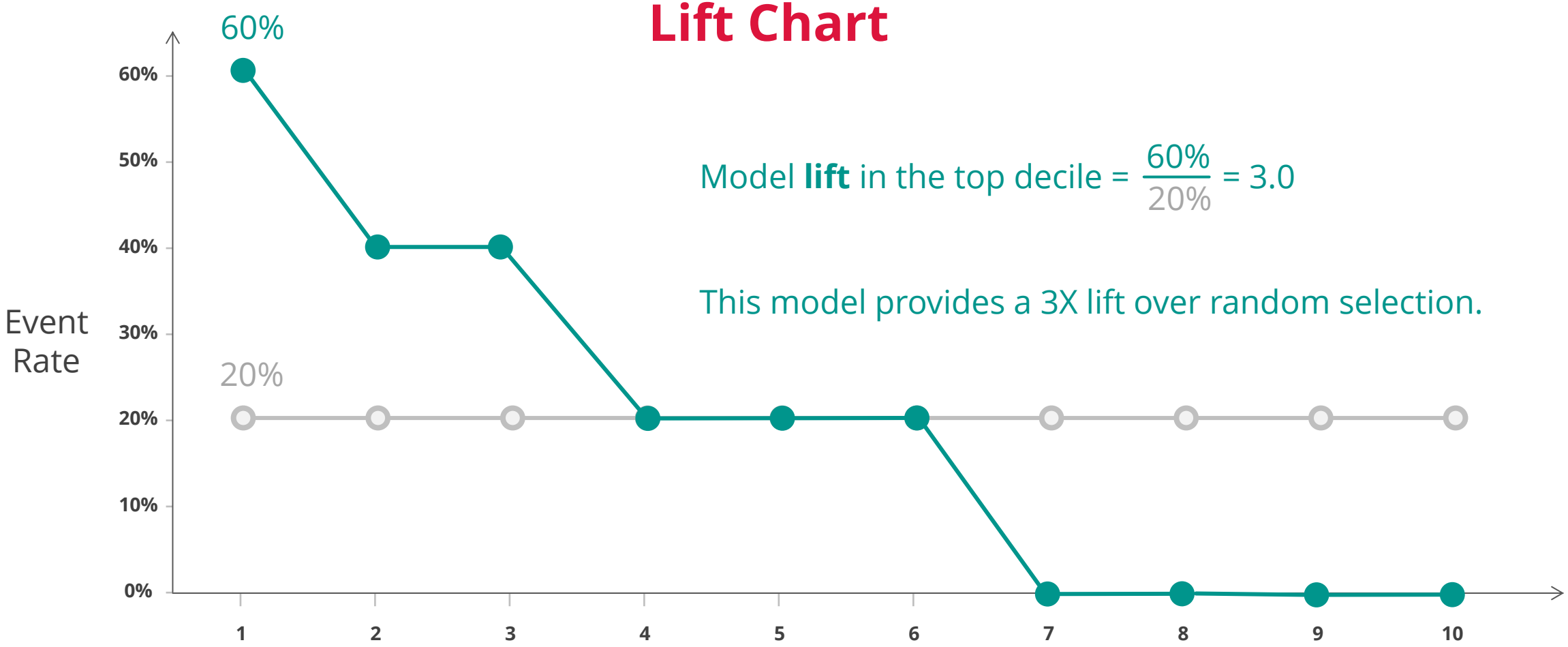
Without having a model, or any other systematic way to identify the targets, let's **randomly** assign each individual into one of the 10 equal groups.





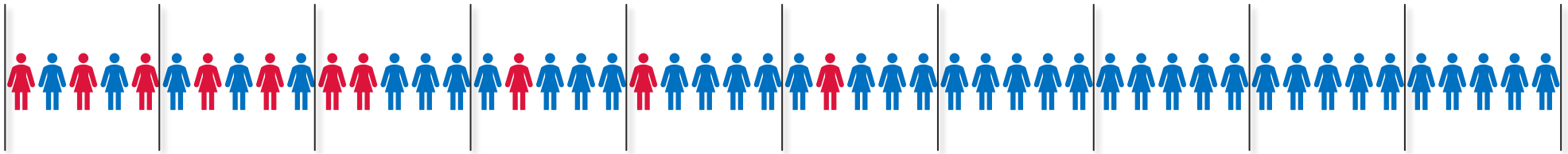
10 equal groups, aka **deciles**

Lift Chart



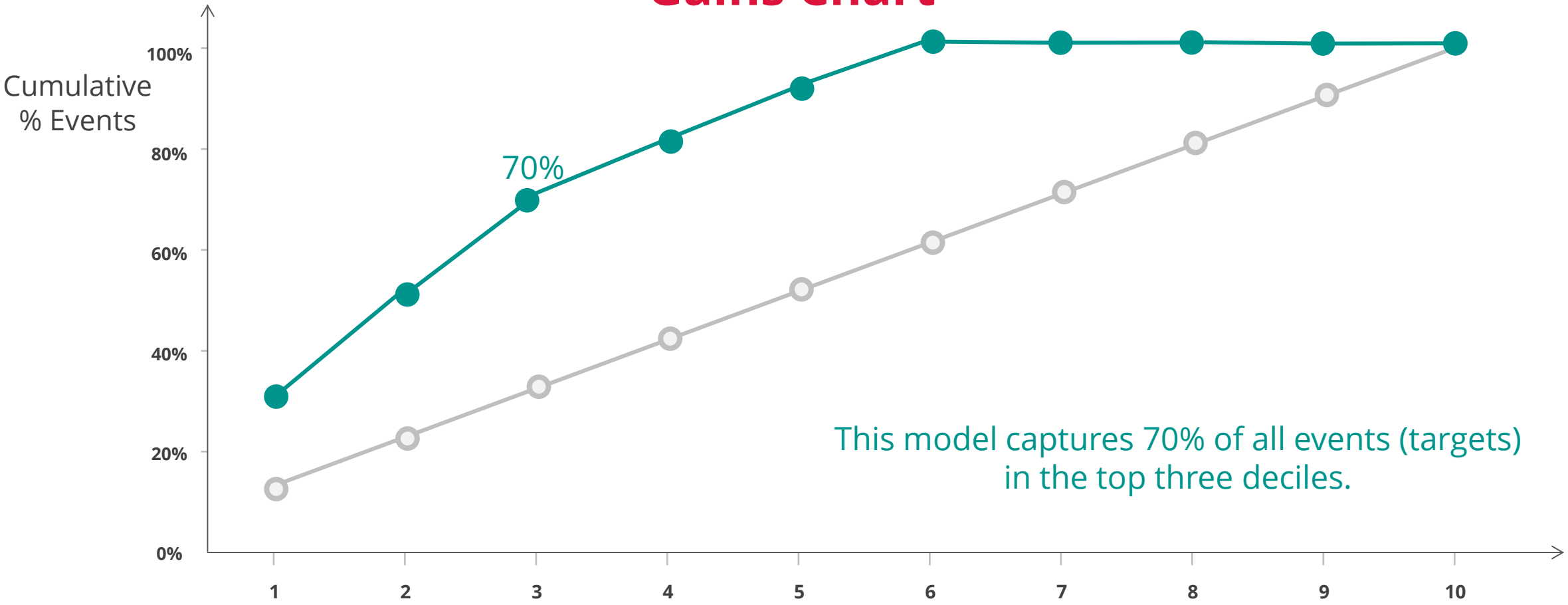
Model **lift** in the top decile = $\frac{60\%}{20\%}$ = 3.0

This model provides a 3X lift over random selection.

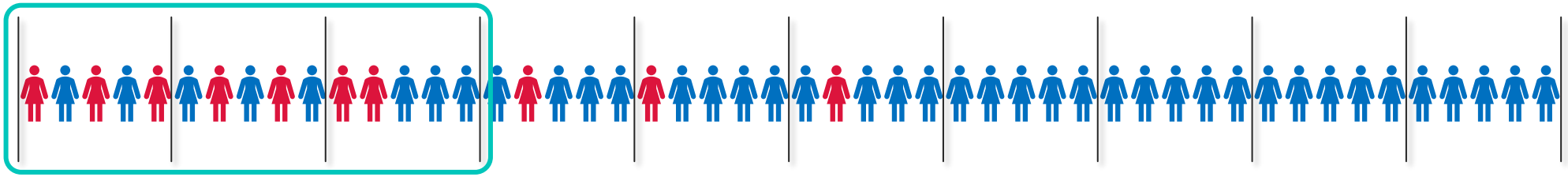


High Probability — Ranked according to **predicted probabilities** (from a model) — Low Probability

Gains Chart



This model captures 70% of all events (targets) in the top three deciles.



High Probability — Ranked according to predicted probabilities (from a model) — Low Probability

When a measure becomes a target,

it ceases to be a good measure.

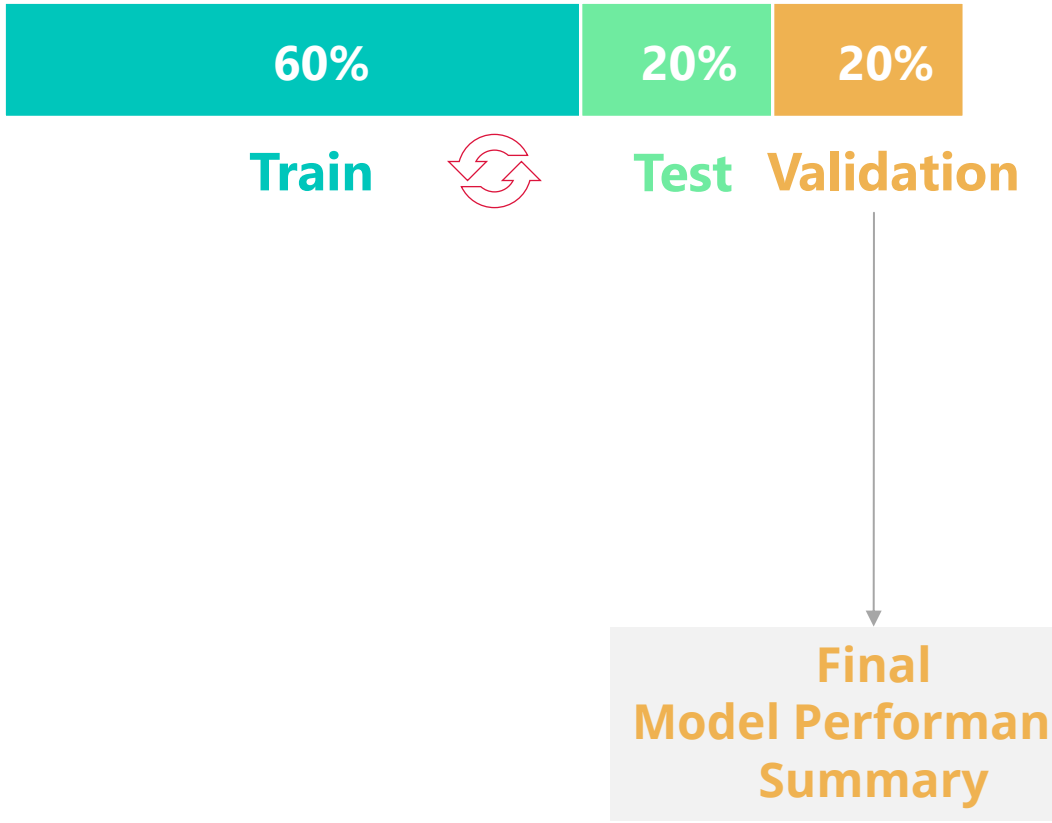
Goodhart's law

Tri-fold Partitioning

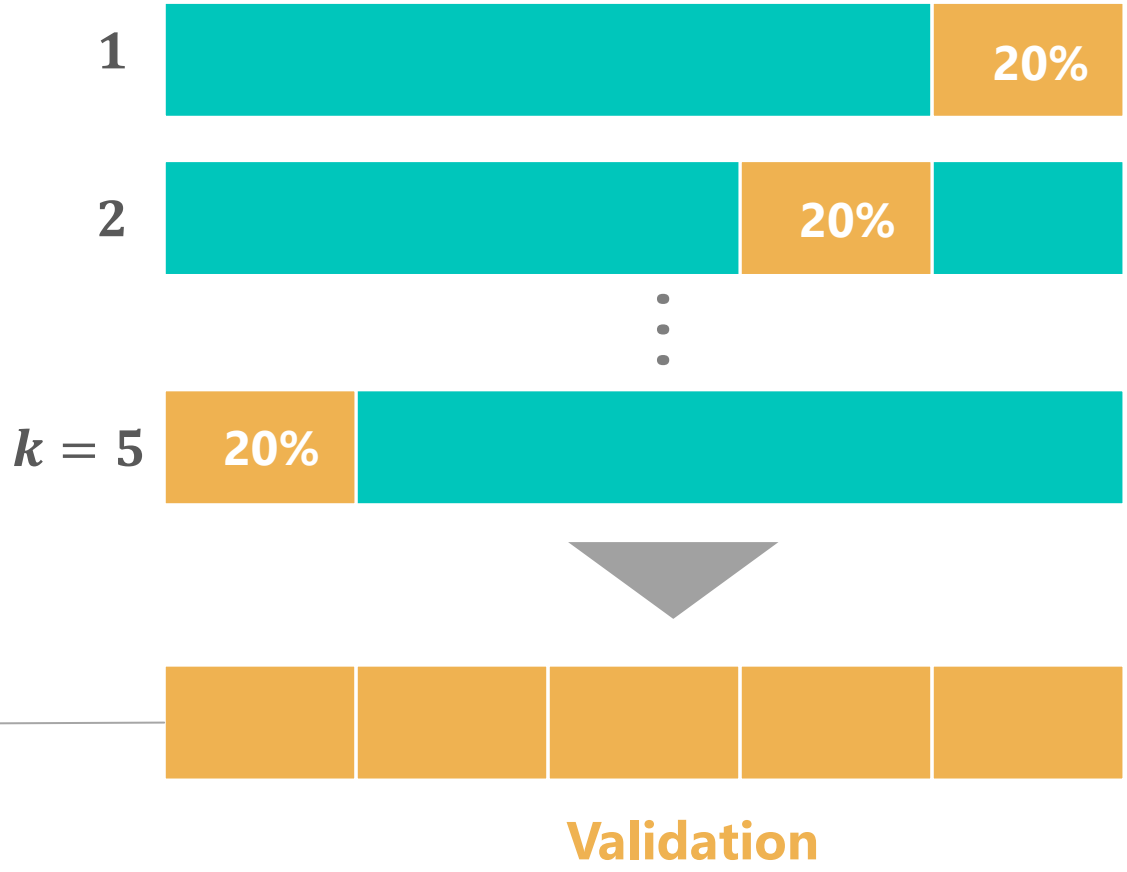
1. Split the modeling dataset into three random partitions: **train**, **test**, and **validation**.
   a) There are no hard and fast rules about the proportions of those three partitions. Suggested proportions are: 60/20/20, 60/30/10, 70/15/15, 70/20/10, or equal split.
   b) If the sample size is small, two-fold partitioning or k-fold cross-validation can be used.
2. **Fit** (train) your models on the **training** set.
3. **Assess** the model accuracy, using one or more metrics, on the **test** set.
   a) Choose metrics that align with the business objectives.
4. Repeat steps 2 and 3 to **refine** your models, e.g., tune hyper-parameters, select a smaller set of predictors.
5. **Select** the best model based on its performance on the **test** set.
6. Once the model is finalized, **measure** the accuracy of the final model using the **validation** (aka the 'hold-out') set.

k-fold Cross Validation

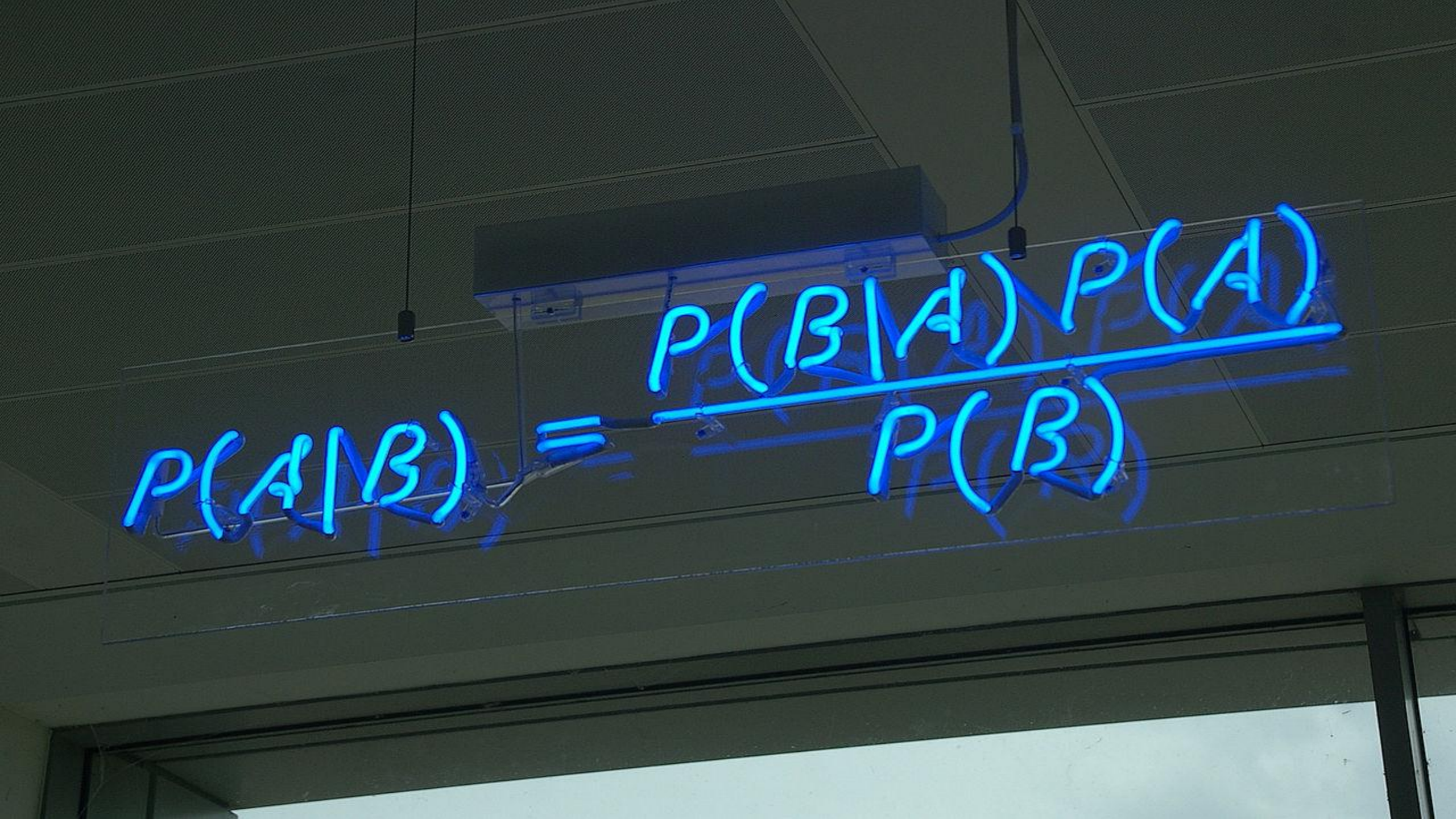
Tri-fold Partitioning



k-fold Partitioning



Bayes' Theorem


P(A|B) = P(B|A) P(A) / P(B)

Thomas Bayes
(1701 – 1761)
English statistician, philosopher, and Presbyterian minister

Bayes' theorem is the law of probability governing the strength of evidence – the rule saying how much to revise our probabilities (change our minds) when we learn a new fact or observe new evidence.

You may want to learn about Bayes' rule if you are:

- A professional who uses statistics,
- A computer programmer working in machine learning;
- A human being.[†]

[†] From Steven Pinker's lecture series on 'Bayesian Reasoning'.

A Bayesian Problem

If a test to detect a disease whose prevalence is 1/1000 has a false positive rate of 5%, what is the chance that a person found to have a positive result has the disease, assuming you know nothing about the person's symptoms or signs?

- Most popular answer: **95%**
- Average answer: **56%**
- Correct answer: **2%** (selected by 18% of doctors!)

Credence
in a hypothesis

Posterior

Is it credible
to begin with?

Prior

Is it more likely to generate
the observed data?

Likelihood

$$p(\text{Hypothesis} \mid \text{Data}) = \frac{p(\text{Hypothesis}) \times p(\text{Data} \mid \text{Hypothesis})}{p(\text{Data})}$$

Marginal

Is the evidence (data) unlikely in general?

Credence in a diagnosis

Is the disease common?

Does the disease usually have those symptoms?

$$p(\text{Disease} \mid \text{Symptoms}) = \frac{p(\text{Disease}) \times p(\text{Symptoms} \mid \text{Disease})}{p(\text{Symptoms})}$$

Are those symptoms unusual across the board?

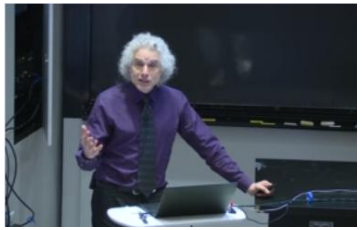
If a test to detect a disease whose prevalence is 1/1000 has a false positive rate of 5%, what is the chance that a person found to have a positive result has the disease, assuming you know nothing about the person's symptoms or signs?

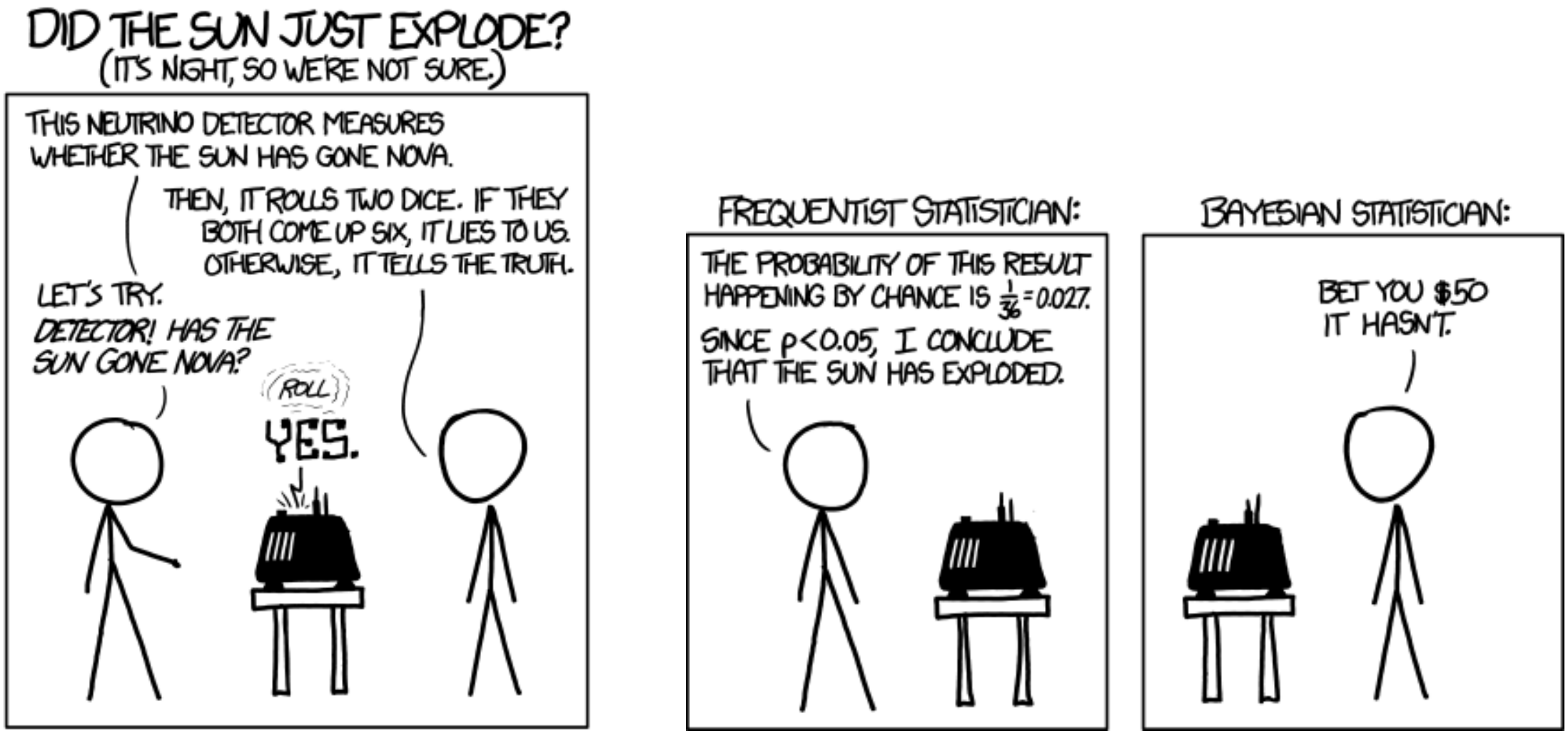
Prevalence = **0.001** Sensitivity = **1.0** (assume)

$$p(\text{Disease} \mid \text{Test positive}) = \frac{p(\text{Disease}) \times p(\text{Test positive} \mid \text{Disease})}{p(\text{Test positive})} = \mathbf{0.0195 \approx 2\%}$$

$$p(\text{Test positive \& Disease}) + p(\text{Test positive \& Healthy})$$

$$(\mathbf{1.0 * 0.001}) + (\mathbf{0.05 * 0.999})$$




DID THE SUN JUST EXPLODE?
(IT'S NIGHT, SO WE'RE NOT SURE.)
THIS NEUTRINO DETECTOR MEASURES WHETHER THE SUN HAS GONE NOVA.
THEN, IT ROLLS TWO DICE. IF THEY BOTH COME UP SIX, IT LIES TO US. OTHERWISE, IT TELLS THE TRUTH.
LET'S TRY. DETECTOR! HAS THE SUN GONE NOVA?
ROLL
YES.
FREQUENTIST STATISTICIAN:
THE PROBABILITY OF THIS RESULT HAPPENING BY CHANCE IS $\frac{1}{36}$=0.027.
SINCE p<0.05, I CONCLUDE THAT THE SUN HAS EXPLODED.
BAYESIAN STATISTICIAN:
BET YOU $50 IT HASN'T.